

1. (a) If $f : R \rightarrow S$ is a homomorphism of rings, show for any $r \in R$ and $n \in \mathbb{Z}$, $f(nr) = nf(r)$.
 (b) Prove that isomorphic rings have the same characteristic. ^{smallest int}
 (c) If $f : R \rightarrow S$ is a homomorphism of rings, must R and S have the same characteristic? ^{st. $\forall a \in R, na = 0$}

(a) pf Case 1. $n \in \mathbb{Z}^+$

$$\text{Then } f(nr) = f(\underbrace{r+r+\dots+r}_{n \text{ times}})$$

$$= \underbrace{f(r) + f(r) + \dots + f(r)}_{n \text{ times}} \text{ since}$$

addition is closed under ring homomorphism

$$= \underline{nf(r)}$$

Case 2. $n = 0$

$$\text{Then } f(nr) = f(0 \cdot r) = f(0_R) = 0_S$$

$$\begin{aligned} &\text{since homomorphism preserve additive identity.} \\ &= 0 \cdot f(r) \\ &= \underline{nf(r)} \end{aligned}$$

Case 3 $n \in \mathbb{Z}^-$

$$\text{Then } f(nr) = f((-n)(-r)) = f(\underbrace{(-r) + (-r) + \dots + (-r)}_{-n \text{ times}})$$

$$= \underbrace{f(-r) + f(-r) + \dots + f(-r)}_{-n \text{ times}}$$

$$= -nf(-r) = -n(-1)f(r) = nf(r)$$

Since the three cases cover all circumstances,
we have proved the statement.

(b) Let R, S be two arbitrary isomorphic rings and φ be an isomorphism from R to S .

Let n be the characteristic of R

$$\text{So } \forall a \in R, na = 0_R$$

$$\text{Since by (a), } \varphi(na) = n\varphi(a)$$

$$\varphi(0_R) = n\varphi(a)$$

Since $\varphi(0_R) = 0_S$ by homomorphism.

$$n\varphi(a) = 0_S$$

So we have proved: for any element a in R , $n\varphi(a) = 0_S$.

And since φ is an isomorphism,

for any element $s \in S$, $\exists$ some element r such that $\varphi(r) = s$

And we have known that $n\varphi(r) = 0_S$

$$\text{So } \underline{\forall s \in S, ns = 0_S}$$

Therefore n is also the characteristic of S .

So we have proved the statement.

(3) R and S do not necessarily have the same characteristic.

This is because when we deduced that $\forall a \in R, n\varphi(a) = 0_S$, we need the surjectivity of φ to ensure any element $s \in S$ is covered. Otherwise, we can have $s \in S$ such that it is not covered so that $\bigcup_{\text{possibly}} ns \neq 0_S$ and n is not the characteristic of

And we can list a counterexample:

$R = \mathbb{Z}_5$ $S = \{0\}$
characteristic = 5 characteristic = 0

but $\varphi: R \rightarrow S$,

sending $z \mapsto 0$ is also a ring homomorphism.

2. Let V be a vector space. Recall that a function $T : V \rightarrow V$ is a *linear transformation* if for all $v, w \in V$ and all $\lambda \in \mathbb{R}$, we have $T(v + w) = T(v) + T(w)$ and $T(\lambda v) = \lambda T(v)$.

- (a) Show that the set of linear transformations from V to V , with usual addition, and *composition of functions* as multiplication, forms a ring.
- (b) Consider the vector space $\mathbb{R}[x]$ and let $\mathcal{L}(\mathbb{R}[x])$ be the ring of linear transformations of $\mathbb{R}[x]$ as defined in the previous part. Consider the element $\frac{d}{dx} \in \mathcal{L}(\mathbb{R}[x])$. Show that there is an element $F \in \mathcal{L}(\mathbb{R}[x])$ such that $\frac{d}{dx} F = 1_{\mathcal{L}(\mathbb{R}[x])}$, but there is no element $G \in \mathcal{L}(\mathbb{R}[x])$ such that $G \frac{d}{dx} = 1_{\mathcal{L}(\mathbb{R}[x])}$.

(a) Denote the set of linear transformations from V to V as $\mathcal{L}(V)$

Let T_1, T_2, T_3 be arbitrary transformations in $\mathcal{L}(V)$

So ① $\forall v \in V, (T_1 + T_2)(v) = T_1(v) + T_2(v)$ is also a linear transformation whose standard matrix is the sum of T_1, T_2 's standard matrices. so $(T_1 + T_2) \in \mathcal{L}(V)$
so $\mathcal{L}(V)$ is closed under addition.

$$\begin{aligned} \text{② } \forall v \in V, (T_1 + T_2)(v) &= T_1(v) + T_2(v) \\ &= T_2(v) + T_1(v) \\ &= (T_2 + T_1)(v) \end{aligned}$$

So addition in $\mathcal{L}(V)$ is commutative

$$\begin{aligned} \text{③ } \forall v \in V, ((T_1 + T_2) + T_3)(v) &= T_1(v) + (T_2(v) + T_3(v)) \\ &= (T_1 + (T_2 + T_3))(v) \end{aligned}$$

So addition in $\mathcal{L}(V)$ is associative

④ Consider $T_0(v) = 0_v$ for all $v \in V$

$$\text{Then } (T_1 + T_0)(v) = (T_0 + T_1)(v) = T_1(v)$$

so $\mathcal{L}(V)$ has additive identity

⑤ for any $T_1(v)$, consider $T'(v) = -T_1(v)$
 which is also a linear transformation
 so $T_1(v) + T'(v) = 0$ for all $v \in V$
 $= T_0(v)$
 so any element in $L(V)$ has
additive inverse

⑥ $\forall v \in V, (T_1 \circ T_2)(v) = T_1(T_2(v))$
 is also a linear transformation
 whose standard matrix is the
 product of the standard matrix
 of T_1 and T_2 , so $(T_1 \circ T_2)(v) \in L(V)$
 so $L(V)$ is closed under
multiplication

⑦ $\forall v \in V, (T_1 \circ T_2) \circ T_3(v) = ((T_1 \circ T_2) \circ T_3)(v)$
 $= T_1(T_2(T_3(v))) = T_1 \circ (T_2 \circ T_3)(v)$
 by the associativity of linear
 transformation
 So $L(V)$ is closed under
 multiplication.

⑧ Consider $T_e(v) = v$.

$$\text{So } \forall v \in V, T_1 \circ T_e(v) = T_1(T_e(v)) \\ = T_1(v)$$

$$T_e \circ T_1(v) = T_e(T_1(v)) \\ = T_e(v)$$

So T_e is a multiplicative identity for $L(V)$

By ①~⑧, $L(V)$ is a ring under the stated addition and multiplication.

(2) ① $f \in L(\mathbb{R}[x])$ such that $\frac{d}{dx} f = I_L(\mathbb{R}[x])$

Consider $F: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ defined by

$$p(x) \mapsto \int_0^x p(t) dt$$

so by Fundamental Theorem of Calculus, $\frac{d}{dx} F(p(x)) = p(x)$.

we have shown in (a) that the $I_L(v) = T(v) = v$.

$$\text{So } \frac{d}{dx} F = I_L(\mathbb{R}[x]).$$

So we have shown the existence of F by example.

(2) Now we prove G (left inverse of $\frac{d}{dx}$ does not exist)

Assume for sake of contradiction that

$\exists G \in L(\mathbb{R}[x])$ such that

$$G \frac{d}{dx} (p(x)) = p(x) \text{ for all } p(x) \in \mathbb{R}[x].$$

Consider $q(x) = c$, so $\frac{d}{dx} q(x) = 0$

and $h(x) = d \neq c$, so $\frac{d}{dx} (ch(x)) = 0$

$$\text{Then } G \left(\frac{d}{dx} q(x) \right) = c$$

$$\Rightarrow G(0) = c$$

$$G \left(\frac{d}{dx} h(x) \right) = d$$

$$\Rightarrow G(0) = d$$

This violating the definition of G as
a function

So contradicts. Such G does not exist.

3. Let d be an integer.

(a) Prove that $\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} \mid a, b \in \mathbb{Z}\}$ is an integral domain.

(b) Show that $\mathbb{Z}_7[\sqrt{3}] = \{a + b\sqrt{3} \mid a, b \in \mathbb{Z}_7\}$ is a field.

(c) Now assume d is also positive and p is a prime. Determine a necessary and sufficient condition for $\mathbb{Z}_p[\sqrt{d}]$ to be a field.

Pf (a) First we prove this is a commutative ring

let $x, y, z \in \mathbb{Z}[\sqrt{d}]$ be arbitrary

$$\begin{aligned} \text{Then } x &= a_1 + b_1\sqrt{d}, \\ y &= a_2 + b_2\sqrt{d}, \\ z &= a_3 + b_3\sqrt{d} \end{aligned} \quad \text{for some } a_1, a_2, a_3, b_1, b_2, b_3 \in \mathbb{Z}$$

$$\text{So 1. } x + y = (a_1 + a_2) + (b_1 + b_2)\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$$

$$\begin{aligned} 2. (x+y) + z &= (a_1 + a_2) + a_3 + [(b_1 + b_2) + b_3]\sqrt{d} \Rightarrow \text{closure under } + \\ &= a_1 + (a_2 + a_3) + b_1 + (b_2 + b_3)\sqrt{d} = x + (y + z) \\ &\Rightarrow \text{associative } + \end{aligned}$$

$$3. x + y = y + x = a_1 + a_2 + (b_1 + b_2)\sqrt{d} \Rightarrow \text{commutative } +$$

$$4. x + 0 = 0 + x \Rightarrow \text{additive identity}$$

$$5. \text{ Since } -a_1 - b_1\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$$

$$\Rightarrow \text{always have additive inverse}$$

$$6. xy = (a_1 + b_1\sqrt{d})(a_2 + b_2\sqrt{d})$$

$$= (a_1 a_2 + b_1 b_2 d) + (a_1 b_2 + a_2 b_1) \sqrt{d} \in \mathbb{Z}[\sqrt{d}]$$

$\Rightarrow$ closure under $\times$

$$\text{Also, } yx = (a_1 a_2 + b_1 b_2 d) + (a_1 b_2 + a_2 b_1) \sqrt{d} = xy$$

since arbitrary $\Rightarrow$ commutative multiplication

$$7. (xy)z = ((a_1 a_2 + b_1 b_2 d) + (a_1 b_2 + a_2 b_1) \sqrt{d}) (a_3 + b_3 \sqrt{d})$$

$$= a_1 a_2 a_3 + b_1 b_2 a_3 d + a_1 b_2 b_3 d + b_1 a_2 b_3 d + (a_1 b_2 a_3 + b_1 a_2 a_3 + a_1 a_2 b_3 + b_1 b_2 b_3 d) \sqrt{d}$$

$$= x(yz) \Rightarrow \text{associative multiplication}$$

$$8. 1x = x1 = x \Rightarrow \text{multiplicative identity,}$$

$$9. x(y+z) = a_1 (a_2 + a_3) + b_1 (b_2 + b_3) \sqrt{d} = a_1 a_2 + a_1 a_3 + (b_1 b_2 + b_1 b_3) \sqrt{d}$$

$$= xy + xz$$

$$(x+y)z = (a_1 + a_2) a_3 + (b_1 + b_2) b_3 \sqrt{d}$$

$$= a_1 a_3 + a_2 a_3 + b_1 b_3 + b_2 b_3 \sqrt{d}$$

By 1~9, $\mathbb{Z}[\sqrt{d}]$ is a commutative ring

$\Rightarrow$ distributive

Now we show that $\mathbb{Z}[\sqrt{d}]$ is a integral domain.

Let x be a nonzero element in $\mathbb{Z}[\sqrt{d}]$

So $x = a + b\sqrt{d}$ for some $a, b \in \mathbb{Z}$ where at least one of $a, b \neq 0$.

Let y be an arbitrary nonzero element in $\mathbb{Z}[\sqrt{d}]$

So $y = m + n\sqrt{d}$ for some $m, n \in \mathbb{Z}$ where at least one of $m, n \neq 0$

Therefore $xy = yx = am + bnd + (an + bm)\sqrt{d} \neq 0$

Consider the worst situation where one of a, b and one of m, n are zero.

If $a = 0, n = 0$, then $bm \neq 0$, so $xy \neq 0$

If $a = 0, m = 0$, then $bnd \neq 0$, so $xy \neq 0$

If $b = 0, n = 0$, then $am \neq 0$, so $xy \neq 0$

If $b = 0, m = 0$, then $an \neq 0$, so $xy \neq 0$

So we have proved $\mathbb{Z}[\sqrt{d}]$ is an integral domain.

(b) Exactly the same as (a) except in modular arithmetic

we can prove $\mathbb{Z}_7[\sqrt{3}]$ is a commutative ring

Now we similarly prove that $\mathbb{Z}_7[\sqrt{3}]$ is a field by proving any nonzero element has a multiplicative inverse.

Let x be an arbitrary element in $\mathbb{Z}_7[\sqrt{3}]$ which is not 0.

So $x = a + b\sqrt{3}$ for some $a, b \in \mathbb{Z}_7$, a and b are not both 0.

Now we let $y = m + n\sqrt{3}$, ^{where} m and $n \in \mathbb{Z}$,

Assume that $xy = [1]$, what we want to do is to solve the equation and conclude that such y always exists.

$$xy = [1] \Rightarrow \begin{cases} am + 3bn = [1]_7, & \textcircled{1} \\ an + bm = [0]_7. & \textcircled{2} \end{cases}$$

Note that $\mathbb{Z}_7$ is a field, since it is a commutative ring, and because 7 is prime, every element in $\mathbb{Z}_7$ has a multiplicative inverse by theorem in Chapter 2.

Since a and b can not both be 0,

Case 1 Assume at least $a \neq 0$,

Then by $\mathbb{Z}_7$ as a field, a^{-1} always exist

Consider $n = -ba^{-1}m$, so $an + bm = -bm + bm$

Then we solve $\textcircled{1}$, which is $= [0]_7$.
 $am = [1]_7 + 3b^2a^{-1}m$ $\textcircled{2}$

Since $\mathbb{Z}_7$ is a field, this also always has a solution, because $am = [1]$ always has a solution ^{for m} and $[1]_7 + 3b^2a^{-1}m$ is some multiple of $[1]_7$. Let $m = [k]_7$ denote the solution for $am = [1]$ and $[1]_7 + 3b^2a^{-1}m = g[1]_7$ we have $m = g[k]_7$ as the solution to $\textcircled{D}$.
Therefore the system always has a solution.

Case 2: Assume at least $b \neq 0$.

Same as case 1, we have $m = -b^{-1}an$ be a solution of $\textcircled{D}$, and then we can always find a solution to $\textcircled{D}$, since $3bn = [1]_7 + b^{-1}a^2n$ always has a solution which is a multiple of $(3b) \cdot n = [1]_7$, guaranteed by $\mathbb{Z}_7$ as a field.

Therefore the system always has a solution.

So we can conclude that any nonzero element in $\mathbb{Z}_7[\sqrt{3}]$ has a multiplicative inverse.

Since $\mathbb{Z}_p[\sqrt{d}]$ is a commutative ring and every nonzero element of it has a multiplicative inverse, it is a field.

(c) The condition is that d^2 is not congruent to 0 (modulo p)

This is because like in (b), we will have to solve for $(d^2 b)n = [1]_p$ when dealing with the case (at least $b \neq 0$), if we denote an arbitrary element π in $\mathbb{Z}_p[\sqrt{d}]$ as $a + b\sqrt{d}$.

If $d^2 \equiv 0 \pmod{p}$, then the equation $[0]_p n = [1]_p$ has no solution. $\Leftrightarrow$ contrapositive,
if the equation has solution, then $d^2 \not\equiv 0 \pmod{p}$

$\Rightarrow$ so $d^2 \not\equiv 0 \pmod{p}$ is necessary condition

If $d^2 \not\equiv 0 \pmod{p}$, then we can always solve the equation like in (b) so that any element in $\mathbb{Z}_p[\sqrt{d}]$ always have a multiplicative inverse.

$\Rightarrow$ so $d^2 \not\equiv 0 \pmod{p}$ is sufficient condition

$\Rightarrow$ Therefore sufficient and necessary.

4. Let R be a commutative ring in which $a^2 = 0$ only if $a = 0$. Show that if $q(x) \in R[x]$ is a zero divisor in $R[x]$, then if:

$$q(x) = \underline{a_0 x^n} + \underline{a_1 x^{n-1}} + \dots + \underline{a_n},$$

there is an element $b \neq 0$ in R such that $ba_0 = ba_1 = \dots = ba_n = 0$.

Proof Assume $q(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$ is a zero divisor in $R[x]$.

So at least one of $a_0, a_1, \dots, a_n$ is not zero and there exists $p(x) = b_0 x^m + b_1 x^{m-1} + \dots + b_m \in R[x]$ $\neq 0$ such that $q(x) \cdot p(x) = 0$ (1)

So

$$\underline{a_0 b_0 x^{m+n} + (a_1 b_0 + a_0 b_1) x^{m+n-1} + (a_0 b_2 + a_1 b_1 + a_2 b_0) x^{m+n-2} + \dots + a_n b_m = 0} \quad (2)$$

Therefore $a_0 b_0 = 0 \Rightarrow b_0$ is a zero divisor in R

* Note that $b_0 \neq 0$ since this is the term with highest degree of $p(x)$, by our assumption.

So $\underline{b_0^{(ER)} \neq 0}$ since $\forall a \in R, a^2 = 0$ iff $a = 0$.

And recursively we have $\underline{b_0^{2k} \neq 0, k \in \mathbb{Z}}$. (3)

(3) implies that all even powers of b_0 are non-zero.

Now we want to show $b_n^{m+1} \neq 0$ no matter it is even or odd powers.

Let b_0^{2k+1} ($k \in \mathbb{Z}$) be an arbitrary odd times of multiples of b_0 .

Assume it is 0 for contradiction.

Then $b_0^{2k+2} = b_0^{2k+1} \cdot b_0 = 0_R b_0 = 0_R$, which contradicts with b_0^{2k} ($k \in \mathbb{Z}$) $\neq 0_R$. So $b_0^{2k+1} \neq 0$.

Hence any multiple of b_0 is non zero, ④

$$\begin{aligned}\text{By ②, we have } \underline{a_1 b_0 + a_0 b_1} &= 0 \\ \Rightarrow b_0 (a_1 b_0 + a_0 b_1) &= 0 \\ \Rightarrow \underline{b_0^2 a_1 = (b_0 a_0) b_1} &= 0\end{aligned}$$

$$\begin{aligned}\text{Also by ②, } \underline{a_0 b_2 + a_1 b_1 + a_2 b_0} &= 0 \\ \Rightarrow b_0^2 (a_0 b_2 + a_1 b_1 + a_2 b_0) &= 0 \\ \Rightarrow \underbrace{b_0^3 a_0}_{=0} + \underbrace{(b_0^2 a_1) b_1}_{=0} + b_0^3 a_2 &= 0 \quad (\text{since } R \text{ is commutative}) \\ \Rightarrow \underline{b_0^3 a_2} &= 0\end{aligned}$$

We can then see the pattern:

In polynomial's term with x^k (starting from $k=0$)

which is $\sum_{\substack{i+j=k \\ i \leq n \\ j \leq m}} a_i b_j$, if we have known that $b_0 a_0 = 0, b_0^2 a_1 = 0, \dots, b_0^k a_{k-1} = 0$

And by multiplying b_0^k on both side, we can get $b_0^{k+1} a_k = 0$ (5)

We can prove (5) by induction on the power of x (k , where $k \geq 0$)

Base case: $k=0, b_0 a_0 = 0$

Inductive step: Assume $b_0 a_0 = 0, b_0^2 a_1 = 0, \dots, b_0^k a_{k-1} = 0$

Then since the term with x^k is

$$\sum_{\substack{i+j=k \\ i \leq n \\ j \leq m}} a_i b_j = 0$$

$$\Rightarrow b_0^k \sum_{\substack{i+j=k \\ i \leq n \\ j \leq m}} a_i b_j = 0$$

$$\Rightarrow b_0^k \cdot b_0 a_k = 0$$

$$\Rightarrow \underline{b_0^{k+1} a_k = 0}$$

Therefore we can conclude: $\forall 0 \leq i \leq n, \underline{b_0^{i+1} \cdot a_i = 0}$ (6)
by the property of $\forall$ or on nty.

Combining ④ (b_m^{n+1} is non-zero) and ⑥
we have finished the proof.

So b_m^{n+1} is the b for the requirement.